Nama : Winawati

NIM : 231331062

Kelas : 2BTTE

Tugas 5 Antena

1. Sebuah antena susun side-by-side terdiri dari 5 elemen antena dipol setengah panjang gelombang (setengah lamda). Jarak antar elemen antena adalah 0.3λ. Adapun arus Listrik yang masuk ke masing-masing elemen adalah $I_1 = 1\angle 0°\ mA, I_2 = 4\angle 15°\ mA$, $I_3 = 6\angle 30°\ mA$, $I_4 = 4\angle 45°\ mA$, dan $I_5 = 1\angle 60°\ mA$. Hitunglah:
   (a) Harga impedansi input pada elemen pertama Z_1
   (b) Harga impedansi input pada elemen keempat Z_4

Diberikan:

- Panjang elemen: $\frac{\lambda}{2}$
- Jarak antar elemen: 0.3λ
- Arus:
  - $I_1 = 1\angle 0°\text{mA}$
  - $I_2 = 4\angle 15°\text{mA}$
  - $I_3 = 6\angle 30°\text{mA}$
  - $I_4 = 4\angle 45°\text{mA}$
  - $I_5 = 1\angle 60°\text{mA}$

Hitung:

(a) Impedansi input Z_1 pada elemen pertama

$$[V] = [Z] \cdot [I]$$

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{bmatrix} = \begin{bmatrix} Z_{11} Z_{12} Z_{13} Z_{14} \\ Z_{21} Z_{22} Z_{23} Z_{24} \\ Z_{31} Z_{32} Z_{33} Z_{34} \\ Z_{41} Z_{42} Z_{43} Z_{44} \\ Z_{51}\ Z_{52}\ Z_{53}\ Z_{54} \end{bmatrix} \cdot \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix}$$

$$Z_{11} = Z_{22} = Z_{33} = Z_{44} = Z_{55} = (73 + j42{,}5)\ \Omega$$

$$V_1 = Z_{11} \cdot I_1 + Z_{12} \cdot I_2 + Z_{13} \cdot I_3 + Z_{14} \cdot I_4 + Z_{15} \cdot I_5$$

Jadi:

$Z_{11} \Rightarrow Z_{11} = (73 + J42{,}5)\Omega$

$Z_{12} \Rightarrow Z_{21} = (30 - j34)\Omega \Rightarrow 45{,}343 \angle -48{,}576^o$

$Z_{13} \Rightarrow Z_{31} = (-21 - j15)\Omega \Rightarrow 25{,}8\angle -144{,}46^o$

$Z_{14} \Rightarrow Z_{41} = (-9 + j19)\Omega \Rightarrow 21{,}023\angle 115{,}346^o$

$Z_{15} \Rightarrow Z_{15} = (18 + j5)\Omega \Rightarrow 18{,}681 \angle 15{,}524^o$

$$V_1 = Z_{11} \cdot I_1 + Z_{12} \cdot I_2 + Z_{13} \cdot I_3 + Z_{14} \cdot I_4 + Z_{15} \cdot I_5$$

$$Z_1 = \frac{V_1}{I_1} = \frac{Z_{11} \cdot I_1 + Z_{12} \cdot I_2 + Z_{13} \cdot I_3 + Z_{14} \cdot I_4 + Z_{15} \cdot I_5}{I_1}$$

$$Z_1 = (73 + J42{,}5)\Omega + (30 - j34)\Omega \cdot \frac{4\angle 15^o}{1\angle 0^o} + (-21 - j15)\Omega \cdot \frac{6\angle 30^o}{1\angle 0^o} + (-9 + j19)\Omega \cdot \frac{4\angle 45^o}{1\angle 0^o} + (18 + j5)\Omega \cdot \frac{1\angle 60^o}{1\angle 0^o}$$

$$Z_1 = (73 + J42{,}5)\Omega + 45{,}343 \angle -48{,}576^o \cdot 4\angle 15^o + 25{,}8\angle -144{,}46^o \cdot 6\angle 30^o + 21{,}023\angle 115{,}346^o \cdot 4\angle 45^o + 18{,}681 \angle 15{,}524^o \cdot 1\angle 60^o$$

$$Z_1 = (73 + J42{,}5)\Omega + 45{,}343 \angle -48{,}576^o \cdot 4\angle 15^o + 25{,}8\angle -144{,}46^o \cdot 6\angle 30^o + 21{,}023\angle 115{,}346^o \cdot 4\angle 45^o + 18{,}681 \angle 15{,}524^o \cdot 1\angle 60^o$$

$$Z_1 = (73 + J42{,}5) + 151{,}16 - j100{,}33 - 64{,}45 - j140{,}68 - 78{,}98 + j28{,}86 + 4.74 + j18.07$$

$$Z_1 = (85{,}47 - j151{,}58)\Omega$$

(b) Impedansi input Z_4 pada elemen keempat

Impedansi input elemen keempat dihitung dengan cara serupa:

$$[V] = [Z] \cdot [I]$$

$$\begin{matrix} V_1 \\ V_2 \\ [V_3] \\ V_4 \\ V_5 \end{matrix} = \begin{matrix} Z_{11}Z_{12}Z_{13}Z_{14} \\ Z_{21}Z_{22}Z_{23}Z_{24} \\ [Z_{31}Z_{32}Z_{33}Z_{34}] \\ Z_{41}Z_{42}Z_{43}Z_{44} \\ Z_{51}\,Z_{52}\,Z_{53}\,Z_{54} \end{matrix} \cdot \begin{matrix} I_1 \\ I_2 \\ [I_3] \\ I_4 \\ I_5 \end{matrix}$$

$$Z_{11} = Z_{22} = Z_{33} = Z_{44} = Z_{55} = (73 + j42{,}5)\ \Omega$$

$$V_4 = Z_{41} \cdot I_1 + Z_{42} \cdot I_2 + Z_{43} \cdot I_3 + Z_{44} \cdot I_4 + Z_{45} \cdot I_5$$

$$Jadi:$$

$$Z_{44} \Rightarrow Z_{44} = (73 + J42{,}5)\Omega$$

$$Z_{41} \Rightarrow Z_{14} = (-9 + j19)\Omega \Rightarrow 21{,}023\angle 115{,}346^o$$

$$Z_{42} \Rightarrow Z_{24} = (-21 - j15)\Omega \Rightarrow 25{,}8\angle -144{,}46^o$$

$$Z_{43} \Rightarrow Z_{34} = (30 - j34)\Omega \Rightarrow 45{,}343\angle -48{,}576^o$$

$$Z_{45} \Rightarrow Z_{54} = (30 - j34)\Omega \Rightarrow 45{,}343\angle -48{,}576^o$$

$$V_4 = Z_{41} \cdot I_1 + Z_{42} \cdot I_2 + Z_{43} \cdot I_3 + Z_{44} \cdot I_4 + Z_{45} \cdot I_5$$

$$Z_4 = \frac{V_4}{I_4} = \frac{Z_{41}.I_1 + Z_{42}.I_2 + Z_{43}.I_3 + Z_{44}.I_4 + Z_{45}.I_5}{I_4}$$

$$Z_4 = 21{,}023\angle 115{,}346° \cdot \frac{1\angle 0°}{4\angle 45°} + 25{,}8\angle -144{,}46° \cdot \frac{4\angle 15°}{4\angle 45°} + 45{,}343\angle -48{,}576° \cdot \frac{6\angle 30°}{4\angle 45°} + (73 + J42{,}5)\Omega + 45{,}343\angle -48{,}576° \cdot \frac{1\angle 60°}{4\angle 45°}$$

$$Z_4 = 5{,}256\angle 70{,}346^o + 25{,}8\angle -174{,}46^o + 68{,}015\angle -63{,}576^o + (73 + J42{,}5)\Omega + 11{,}336\angle -33{,}576^o$$

$$Z_4 = 5{,}256\angle 70{,}346^o + 25{,}8\angle -174{,}46^o + 68{,}015\angle -63{,}576^o + (73 + J42{,}5)\Omega + 11{,}336\angle -33{,}576^o$$

$$Z_4 = 1{,}77 + j4{,}94 - 25{,}66 - j2{,}8 + 30{,}58 - j60{,}77 + 73 + j42{,}5 + 9{,}44 - j6{,}28$$

$$Z_4 = 89{,}13 - j22{,}41\ \Omega$$

2. Sebuah antena susun kolinier terdiri dari 5 elemen antena dipol setengah panjang gelombang (setengah lamda). Jarak antar elemen antena adalah 0.3λ. Adapun arus Listrik yang masuk ke masing-masing elemen adalah $I_1 = 1\angle 0°\ mA, I_2 = 1\angle 25°\ mA$, $I_3 = 1\angle 50°\ mA$, $I_4 = 1\angle 75°\ mA$, dan $I_5 = 1\angle 100°\ mA$. Hitunglah:
   (a) Harga impedansi input pada elemen pertama Z_1
   (b) Harga impedansi input pada elemen keempat Z_4

(a) Impedansi input Z_1 pada elemen pertama

$$[V] = [Z] \cdot [I]$$

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} Z_{11}Z_{12}Z_{13}Z_{14} \\ Z_{21}Z_{22}Z_{23}Z_{24} \\ Z_{31}Z_{32}Z_{33}Z_{34} \\ Z_{41}Z_{42}Z_{43}Z_{44} \end{bmatrix} \cdot \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix}$$

$$Z_{11} = Z_{22} = Z_{33} = Z_{44} = (73 + j42{,}5)\ \Omega$$

$$V_1 = Z_{11} \cdot I_1 + Z_{12} \cdot I_2 + Z_{13} \cdot I_3 + Z_{14} \cdot I_4$$

$$Jadi:$$

$$Z_{11} \Rightarrow Z_{11} = (73 + J42{,}5)\Omega$$

$$Z_{12} \Rightarrow Z_{21} = (-0{,}5 - j7)\Omega \Rightarrow 7{,}017\angle -94{,}085^o$$

$$Z_{13} \Rightarrow Z_{31} = (2 - j)\Omega \Rightarrow \ 2{,}236\angle -26{,}565^o$$

$$Z_{14} \Rightarrow Z_{41} = (0{,}5 + 0{,}5j)\Omega \ \Rightarrow \ 0{,}707\angle 45^o$$

$$V_1 = Z_{11} \cdot I_1 + Z_{12} \cdot I_2 + Z_{13} \cdot I_3 + Z_{14} \cdot I_4$$

$$Z_1 = \frac{V_1}{I_1} = \frac{Z_{11} \cdot I_1 + Z_{12} \cdot I_2 + Z_{13} \cdot I_3 + Z_{14} \cdot I_4}{I_1}$$

$$Z_1 = (73 + J42{,}5)\Omega + (-0{,}5 - j7)\ \frac{1\angle 25^o}{1\angle 0^o} + (2 - j) \cdot \frac{1\angle 50^o}{1\angle 0^o} + (0{,}5 + 0{,}5j) \cdot \frac{1\angle 75^o}{1\angle 0^o}$$

$$Z_1 = (73 + J42{,}5)\Omega + (-0{,}5 - j7) \cdot 1\angle 25^o + (2 - j) \cdot 1\angle 50^o + (0{,}5 + 0{,}5j) \cdot 1\angle 75^o$$

$$Z_1 = (73 + J42{,}5)\Omega + 7{,}017\angle -94{,}085^o \cdot 1\angle 25^o + 2{,}236\angle -26{,}565^o \cdot 1\angle 50^o + 0{,}707\angle 45^o \cdot 1\angle 75^o$$

$$Z_1 = (73 + J42{,}5)\Omega + 7{,}017\angle -69{,}085^o + 2{,}236\angle 23{,}435^o + 0{,}707\angle 120^o$$

$$Z_1 = (73 + J42{,}5)\Omega + 2{,}47 - j6{,}57 + 2{,}05 + j0{,}89 - 0{,}354 + j0{,}612$$

$$Z_1 = (77{,}17 + j37{,}43)\Omega$$

(b) Impedansi input Z_4 pada elemen keempat

$$[V] = [Z] \cdot [I]$$

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} Z_{11}Z_{12}Z_{13}Z_{14} \\ Z_{21}Z_{22}Z_{23}Z_{24} \\ Z_{31}Z_{32}Z_{33}Z_{34} \\ Z_{41}Z_{42}Z_{43}Z_{44} \end{bmatrix} \cdot \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix}$$

$$Z_{11} = Z_{22} = Z_{33} = Z_{44} = (73 + j42{,}5)\ \Omega$$

$$V_4 = Z_{41} \cdot I_1 + Z_{42} \cdot I_2 + Z_{43} \cdot I_3 + Z_{44} \cdot I_4$$

$$Jadi:$$

$$Z_{44} \Rightarrow Z_{44} = (73 + J42{,}5)\Omega$$

$$Z_{41} \Rightarrow Z_{14} = (0{,}5 + 0{,}5j)\Omega \Rightarrow 0{,}707\angle 45^o$$

$$Z_{42} \Rightarrow Z_{24} = (2 - j)\Omega \Rightarrow 2{,}236\angle -26{,}565^o$$

$$Z_{43} \Rightarrow Z_{34} = (-0{,}5 - j7)\Omega \Rightarrow 7{,}017\angle -94{,}085^o$$

$$V_4 = Z_{41} \cdot I_1 + Z_{42} \cdot I_2 + Z_{43} \cdot I_3 + Z_{44} \cdot I_4$$

$$Z_4 = \frac{V_4}{I_4} = \frac{Z_{41}.I_1 + Z_{42}.I_2 + Z_{43}.I_3 + Z_{44}.I_4}{I_4}$$

$$Z_4 = (0{,}5 + 0{,}5j) \cdot 1\angle -75^o + (2 - j) \cdot 1\angle -50^o + (-0{,}5 - j7) \cdot 1\angle -25^o + (73 + J42{,}5)\Omega$$

$$Z_4 = 0{,}707\angle 45^o \cdot 1\angle -75^o + 2{,}236\angle -26{,}565^o \cdot 1\angle -50^o + 7{,}017\angle -94{,}085^o \cdot 1\angle -25^o + (73 + J42{,}5)\Omega$$

$$Z_4 = 0{,}707\angle -30^o + 2{,}236\angle -76{,}565^o + 7{,}017\angle -119{,}085^o + (73 + J42{,}5)\Omega$$

$$Z_4 = 0.612 - j0.354 + 0.523 - j2.173 + -3.417 - j6.127 + (73 + J42{,}5)\Omega$$

$$Z_4 = 70{,}72 + j33{,}85\ \Omega$$